# Deutsch Algorithm

## Introduce

- Proposed by David Deutsch in 1985
- The first algorithm to demonstrate the superiority of quantum computers over classical computers.
- The "first true quantum algorithm."

## Problem

- determining whether a black-box function is constant or balanced

| costant | balance |
| --- | --- |
| f(x=0)=0<br>f(x=1)=0 | f(x=0)=0<br>f(x=1)=1 |
| f(x=0)=1<br>f(x=1)=1 | f(x=0)=1<br>f(x=1)=0 |

## Conclusion

- classic computing needs 2 query, quantum computing needs 1 query so quantum computing win.

## Development

- Deutsch Algorithm 1985 → Deutsch-Jozsa Algorithm 1992
    - single-bit inputs → bit strings of arbitrary length.

## Prerequisite (Vocabulary and Knowledge)

- Unitary transformation

- $\oplus$ (modulo 2addition) (XOR operation)
- Register
- Matrix calculation rules
- Describe the state of vectors in Bloch ball

## Practice Description

- Using McGurk effect to show inner logic of Deutsch Algorithm. If we listen, there are two results, but if we watch, then will be four results. What Deutsch Algorithm fixes is the situation we only want to listen.
- 4 random probabilities triggering by bang ,mapping to 4 quantum circuits presents 4 Unitary transformation inside blackbox.
- Your brain tricks you into matching different lip shapes in the video, imagining the same audio as a different pronunciation

## Refrences

[1] C. Bernhardt, *Quantum Computing for Everyone*. Cambridge, MA, USA: The MIT Press, 2020. [Online]. Available: https://doi.org/10.7551/mitpress/11860.001.0001 p145

[2] A. Flarend and R. C. Hilborn, *Quantum Computing: From Alice to Bob*. Oxford, United Kingdom: Oxford Univ. Press, 2022. [Online]. Available: https://doi.org/10.1093/oso/9780192857972.001.0001 p160

[3] SPINQ, *Gemini Lab Experiment Manual*. 2025.

[4] Qiskit Community, *qiskit-textbook*. 2020. [Online]. Available: https://github.com/qiskit-community/qiskit-textbook